

Design and Structural Optimization of a Bike Frame for Bike-Share Deployment

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Bike Frame Design Project Report
ANSYS Simulation

1. Introduction

This project focuses on the design of an electric bicycle frame intended for large-scale deployment in bike-share programs and consumer markets. The ANSYS project focuses on designing a frame that can be (1)lightweight, (2)safe, (3)durable, (4)inexpensive, and (5)easy to use and deploy.



1.1 Impact

The goal of this project is not only to create a functional e-bike frame, but also to consider how the design affects public health, safety, environmental impact, cost, and accessibility.

From a societal perspective, a reliable e-bike frame must be safe but also lightweight and affordable. While the frame must support the rider and road-induced loads without failure, the design should also avoid unnecessary mass and cost so that it remains practical for bike-share deployment. Therefore, this project is not only a structural analysis problem, but also a design tradeoff between accessibility, safety, durability, weight, and cost.

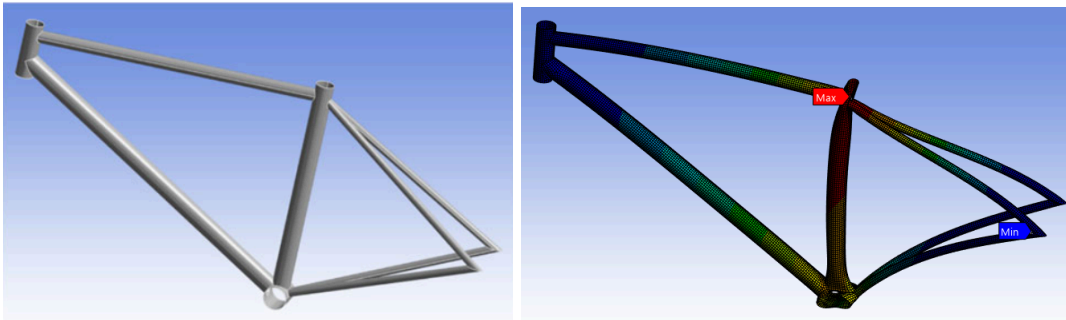
A well-designed e-bike frame can improve transportation access. E-bikes can reduce car use for short trips, lowering emissions and traffic congestion while supporting healthier travel habits. At Cornell, an e-bike share system could help students travel across campus more efficiently and reduce mobility barriers caused by the hills and spread-out campus layout.

1.2 Project Overview

The objective of this project was to analyze an initial bicycle frame geometry and then create a modified e-bike frame design suitable for bike share use. This project consisted of three main parts.

1. Create baseline shell frame and do Initial analysis

First, an initial bicycle frame geometry was analyzed in Ansys with a shell finite element model, with aluminum alloy material properties. The original design was analyzed to understand total deformation, equivalent stress, and safety factor.

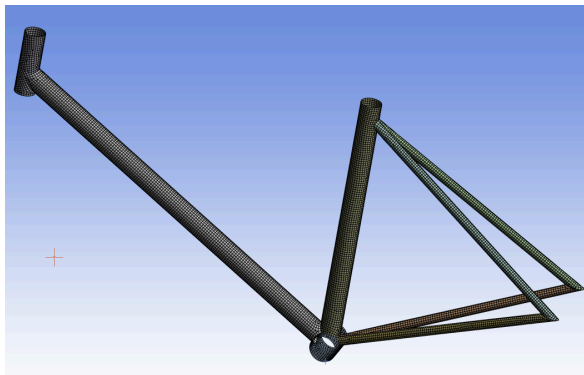


2. Update geometry and analyze change

Second, after analyzing the provided geometry, a new frame design was created. The new design removed the top tube for a more accessible step-through frame, at the cost of weakening the traditional triangular structure of the bike frame. It also added the weight of the battery in the front lower frame. The new design was re-analyzed carefully using finite element analysis.

3. Parametric optimization under SF for mass reduction

Then, a parametric optimization was performed to determine an appropriate tube thickness, balancing mass reduction against the need to maintain a safety factor higher than 2. The thickness was varied from 2mm-4mm range on each pipe to find the optimal combination.



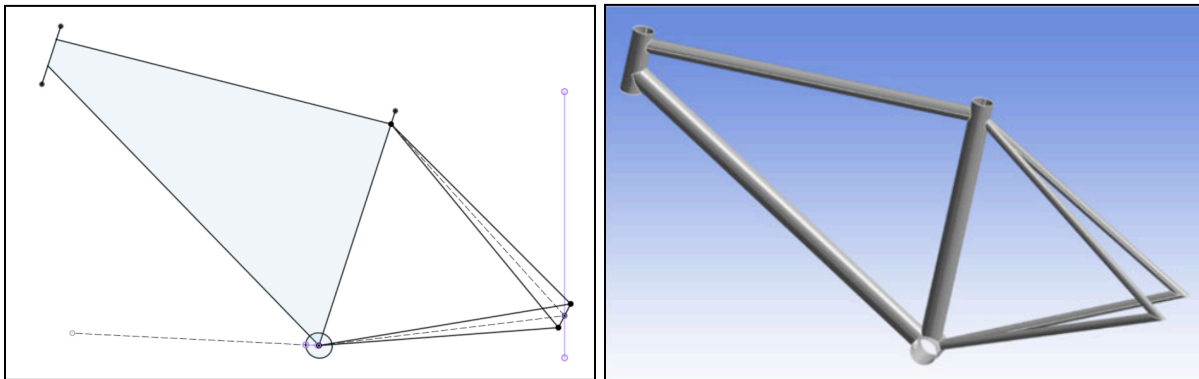
	A	B
1		Enabled
2	✓ Design of Experiments	
3	Input Parameters	
4	Static Structural (B1)	
5	P1 - Body4 Thickness	✓
6	Output Parameters	
7	Static Structural (B1)	
8	P2 - Geometry Mass	
9	P3 - Safety Factor Minimum	
10	Charts	
11	✓ Parameters Parallel	
12	✓ Design Points vs Parameter	

2. Analysis of Provided Geometry

2.1 Initial Geometry Description

- Original bike frame
- Consisted of main frame tubes
- Baseline model used to analyze deformation, stress, and safety factor

The initial geometry was a conventional bicycle frame consisting of the main tubes, including the top tube, down tube, seat tube, chain stays, and seat stays. This geometry served as the baseline model for understanding the structural behavior of a bike frame under simplified loading and boundary conditions. The purpose of this first analysis was to establish a baseline for deformation, stress, and safety factor before modifying the design into an e-bike bike-share frame.



2.2 Mathematical Model (shells)

- Shell model
- Aluminum alloy material
- Initial thickness, element size
- Assumptions

The bicycle frame was modeled using shell elements. This is appropriate because bicycle frame tubes are thin-walled structures, where the tube wall thickness is much smaller than the overall tube diameter and length. Instead of modeling the full 3D solid thickness of every tube, shell elements represent the tube surfaces with an assigned wall thickness.

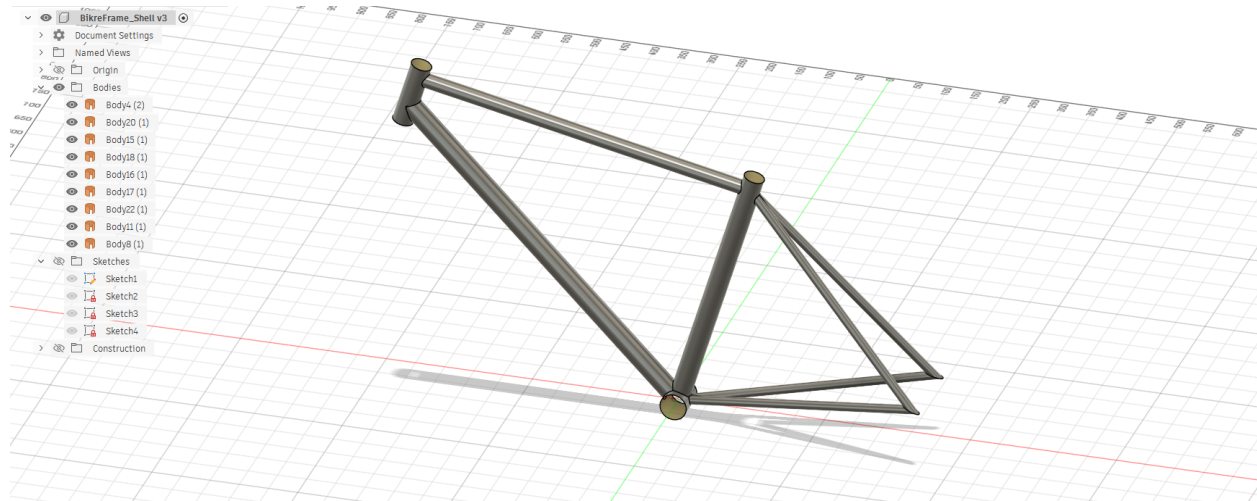
Model Feature	Input
Material	Aluminum alloy from Ansys database
Element type	Shell element
Tube Thickness	2mm
Element size	4mm

Main results evaluated

Total deformation, equivalent stress, safety factor

The simplified structural model assumes that the frame behaves elastically under the applied loading. The analysis uses finite element methods to approximate the displacement field, then calculates strain, stress, and safety factor from the numerical displacement solution.

Shell was made as below:



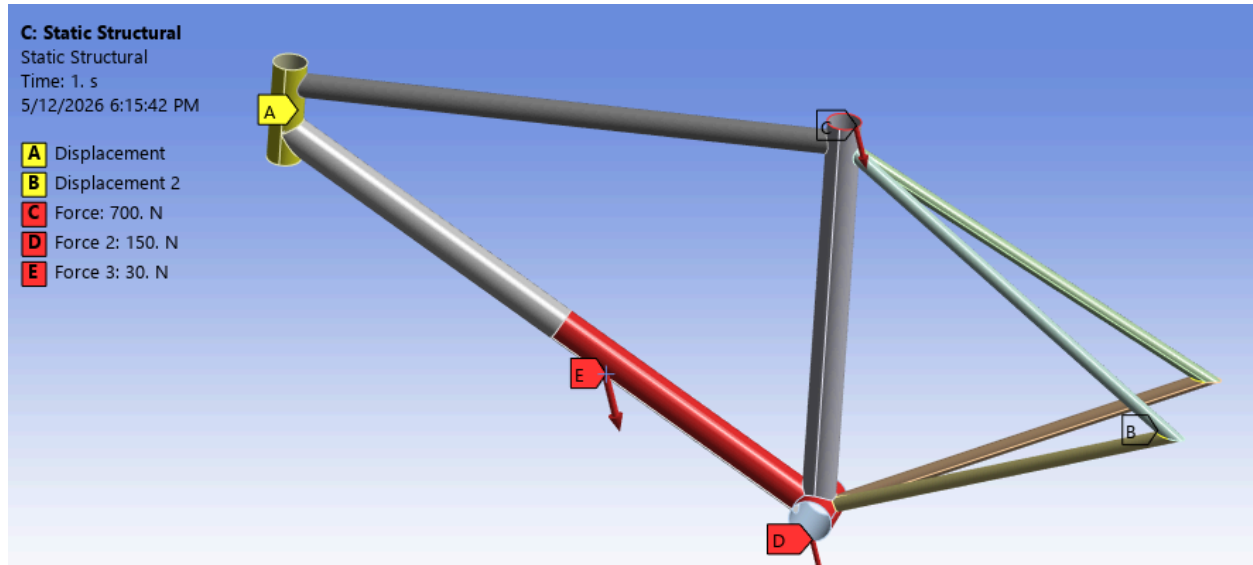
Mesh was made as below:



Statistics	
☐ Nodes	16826
☐ Elements	16832

2.3 Boundary Conditions and Loading

- Supports/constraints
- Applied forces
- Load locations
- Purpose of loading setup



The model included constraints at the wheel/contact or frame support regions and applied forces representing rider and frame loading. Based on the slide image for the shell model setup, forces were applied at several frame locations, including a larger downward load near the seat region and additional loading near other structural points.

The loading set up in the model was:

Load/Boundary Condition	Location	Input
Force (battery)	Mid section of lower bar	30N (~3kg)
Displacement	Front end of bike frame	Fixed end
Displacement	Back end of bike frame	Fixed end
Force (rider weight)	Top of sitting bar	-700N z direction
Force	Bottom of sitting bar	-150N z direction

The purpose of these boundary conditions was to approximate how the bicycle frame carries rider and component loads through the frame tubes into the wheel support regions.

2.4 Numerical Solution Strategy

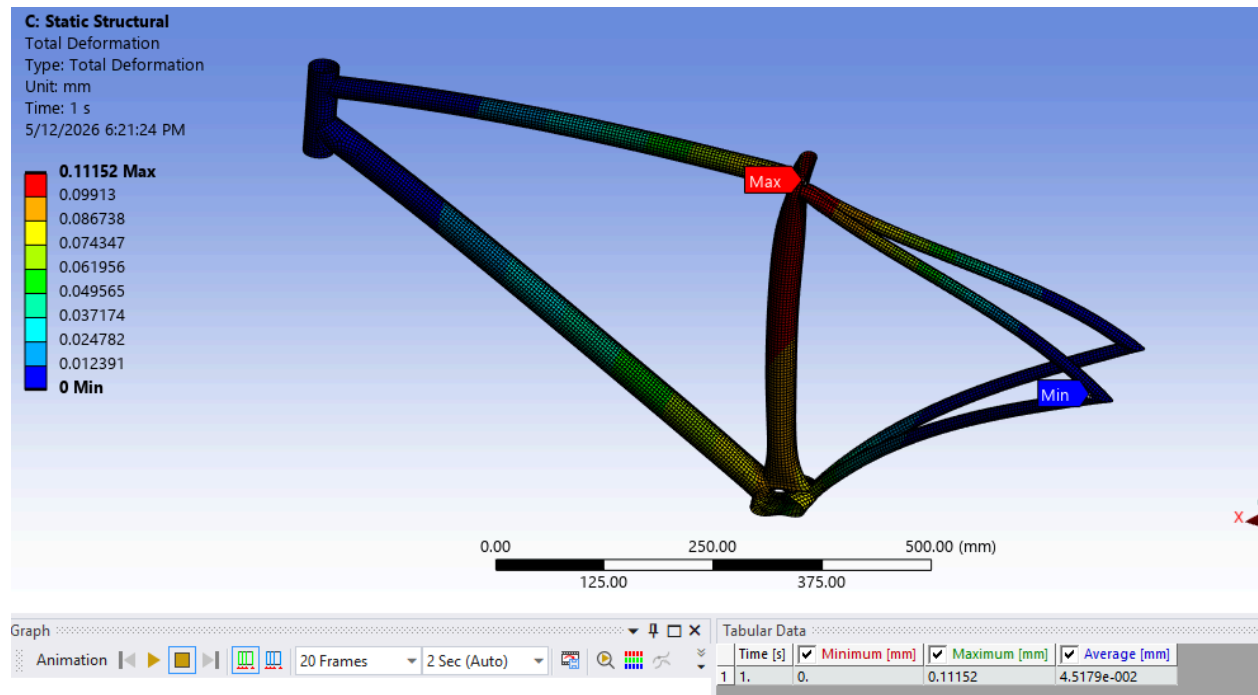
The frame was analyzed using the finite element method in Ansys Mechanical. The geometry was divided into small shell elements, and each element approximated the structural response of a small region of the frame.

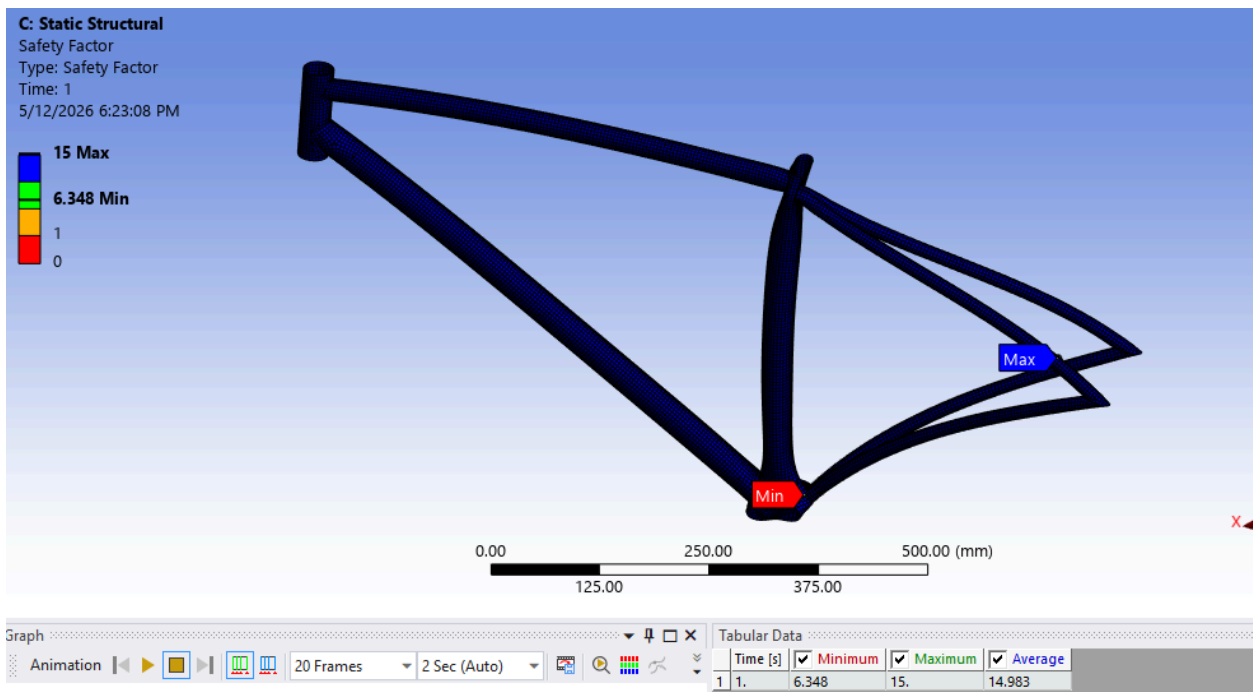
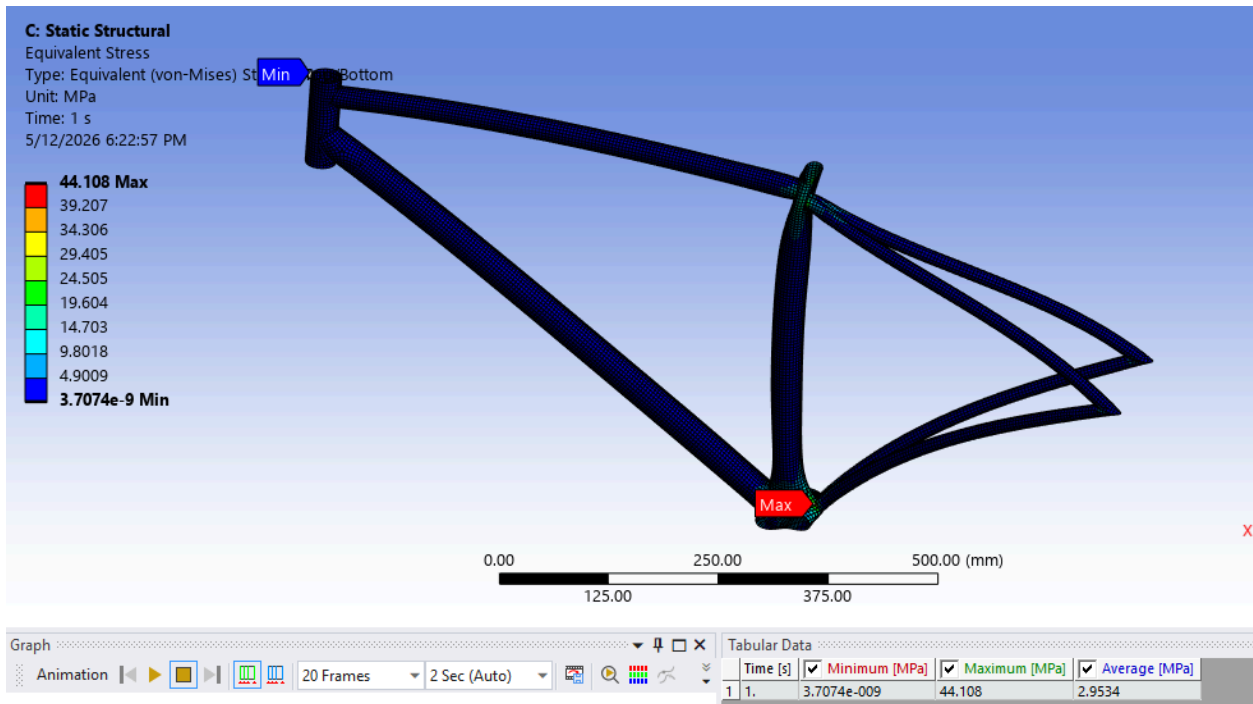
4mm mesh: 16,826 nodes and 16,832 elements. Since shell elements have 6 degrees of freedom per node, the approximate total number of nodal degrees of freedom was: $16826 \times 6 = 100956$ DOF

2.5 Ansys Implementation and Results

- Main stress location: Near the load transfer region at the top
- SF remained well above the required safety factor threshold

Result	Value
Total Deformation	Max 0.111 mm
Equivalent Stress	Max 44.1 MPa
Safety Factor	Min 6.35, Max 15, Avg: 14.98

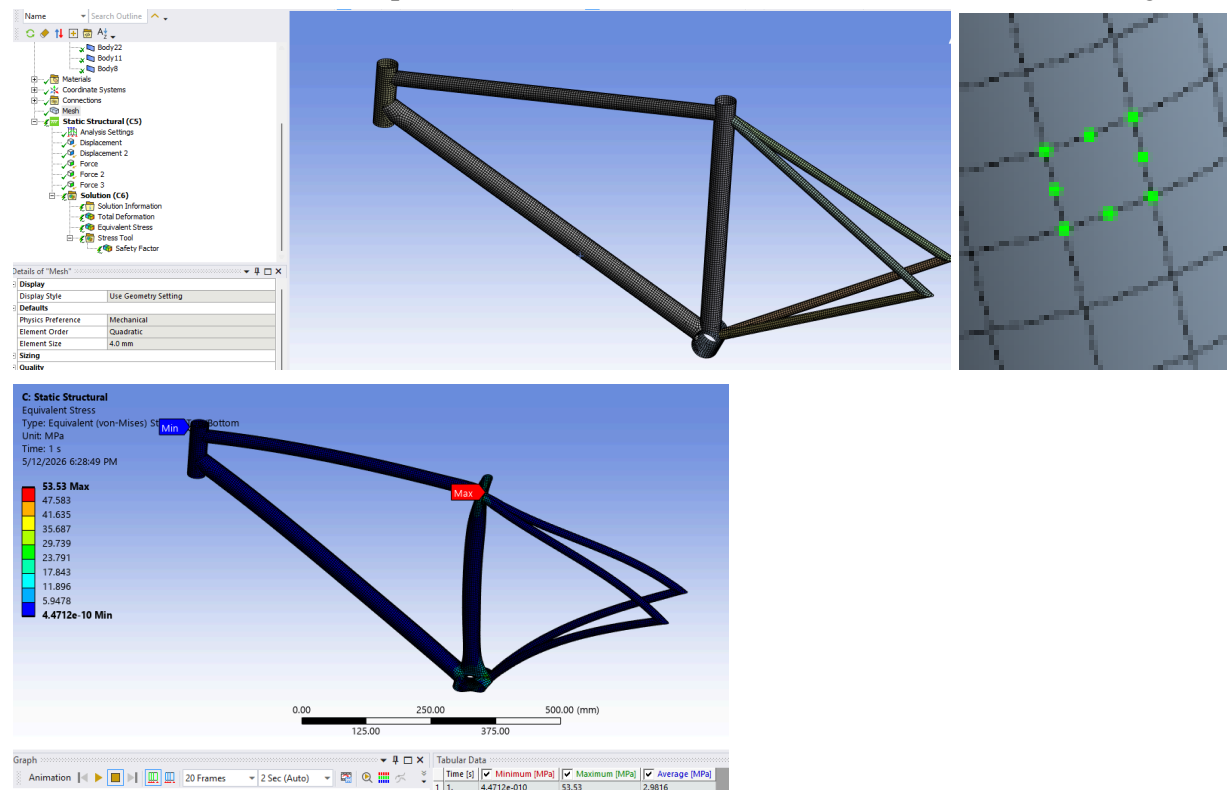




2.6 Verification

Verification Check	Result	Interpretation
Material check	Aluminum alloy from ANSYS database	Matches project assumptions
Boundary condition check	Fixed displacement in x, y, z at required points	Correct essential BC setup
Element order check	Deformation stable, stress changed by ~20.45%	Stress is more interpolation-sensitive
Mesh refinement	Deformation converged strongly; stress changed moderately	Results are reasonable, but stress needs more attention than displacement
Yield comparison	Stress far below ~275 MPa yield stress	Frame remains structurally safe

Then the mesh was switched to **quadratic element order** to see the effect of element order change:



Result	Value
Total Deformation	Max 0.111 mm
Equivalent Stress	Max 53.5 MPa (~20.5% increase)
Safety Factor	Min 5.23, Max: 15, Average: 14.975

- Deformation was similar but stress increased by 20.45%
- Same main stress location and SF still above the required threshold
- → Need to be careful when looking at stress values to predict real life stress!

3. New Design

3.1 New Design Goal and Hypothesis

The goal of the new design was to create a bike-share e-bike frame that is easier to use than a traditional frame. In a public bike-share system, users may have different heights, mobility levels, clothing, backpacks, or bags, so the frame should be easy to mount and dismount quickly. The main design change was removing the top tube to create a step-through frame. This improves accessibility because riders can mount and dismount more easily.

However, removing the top tube also weakens the traditional triangular load path. Therefore, the design needed to maintain enough stiffness and strength through the remaining tubes while also carrying the additional 3 kg battery weight.



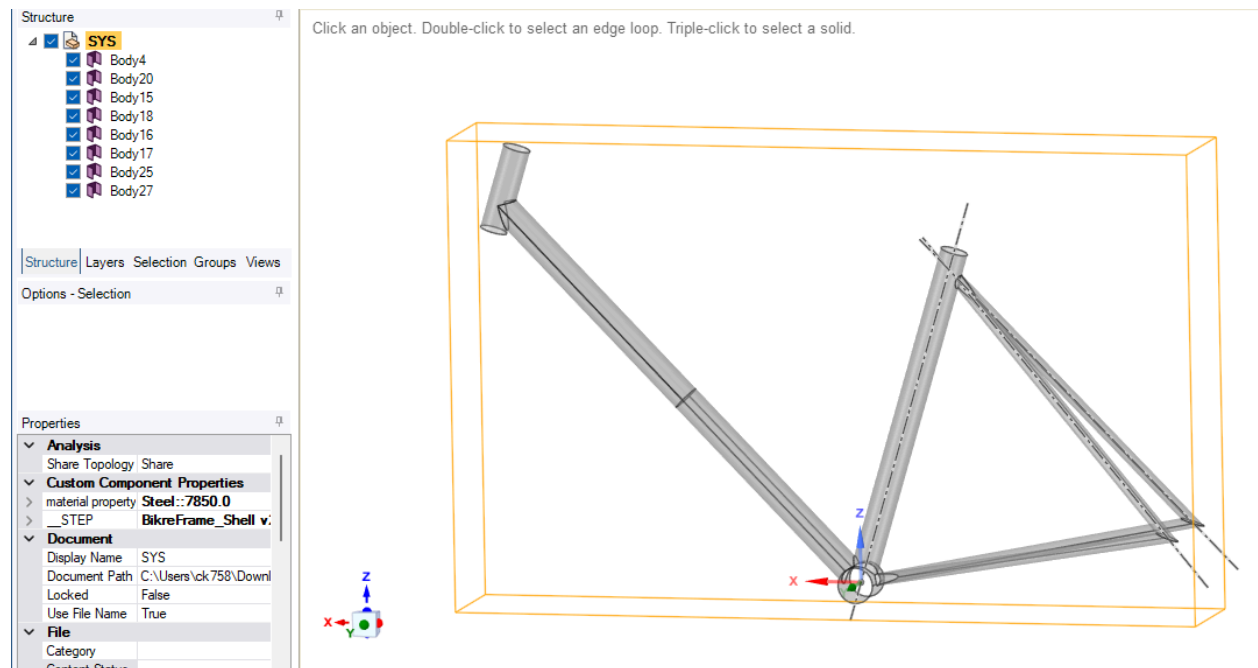
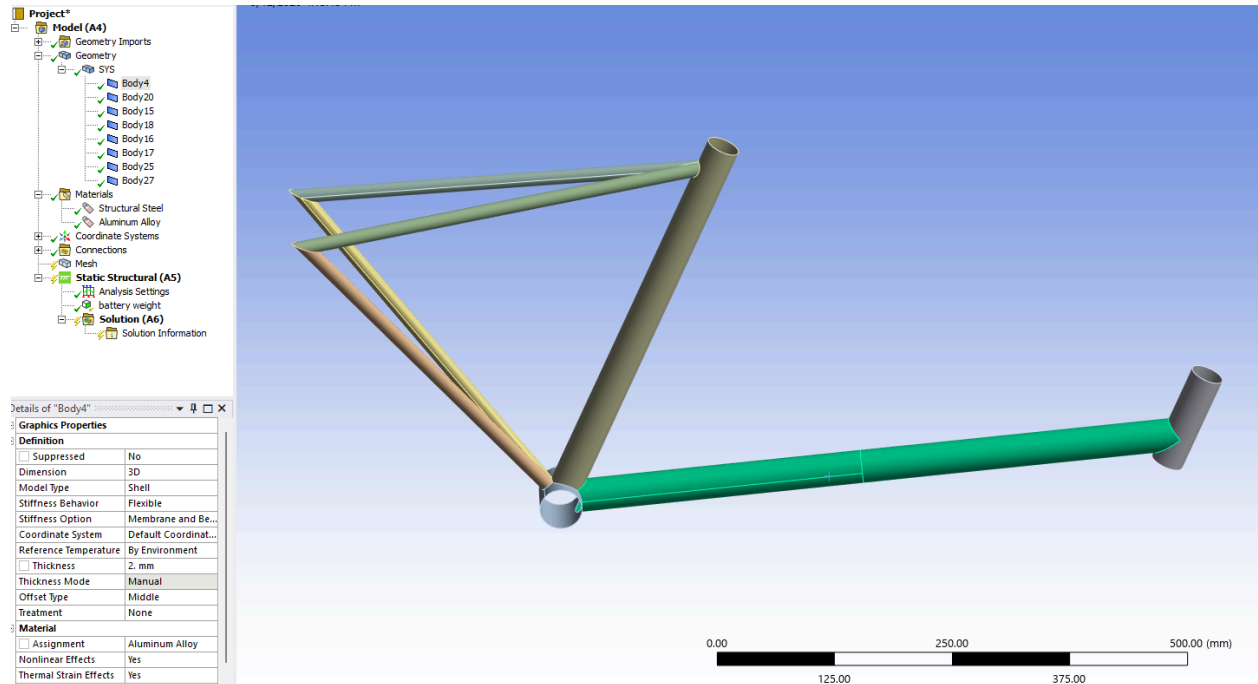
3.2 Design Hypothesis

Removing the top tube would increase deformation and equivalent stress because the frame loses part of its triangular structure. The highest stress will occur near major load-transfer regions, especially around the bottom bracket, seat tube, and lower tube connections. The expected minimum safety factor will decrease compared with the original frame, but remain above the required safety factor of 2 if the lower-frame members were sized appropriately.

3.3 New Geometry Description

- Removed top tube
- Battery weight added

The top tube was removed and battery weight was considered and 30N was added. Shared topology was used so the connected tube members behaved as a continuous structural frame while still allowing separate tube thicknesses to be assigned to different members. This was important because the optimization required different tube thicknesses to be treated as design parameters.



3.4 Ansys Implementation for New Design

- Mesh setup
- Same boundary conditions and load
- Battery weight added

Mesh was as below:

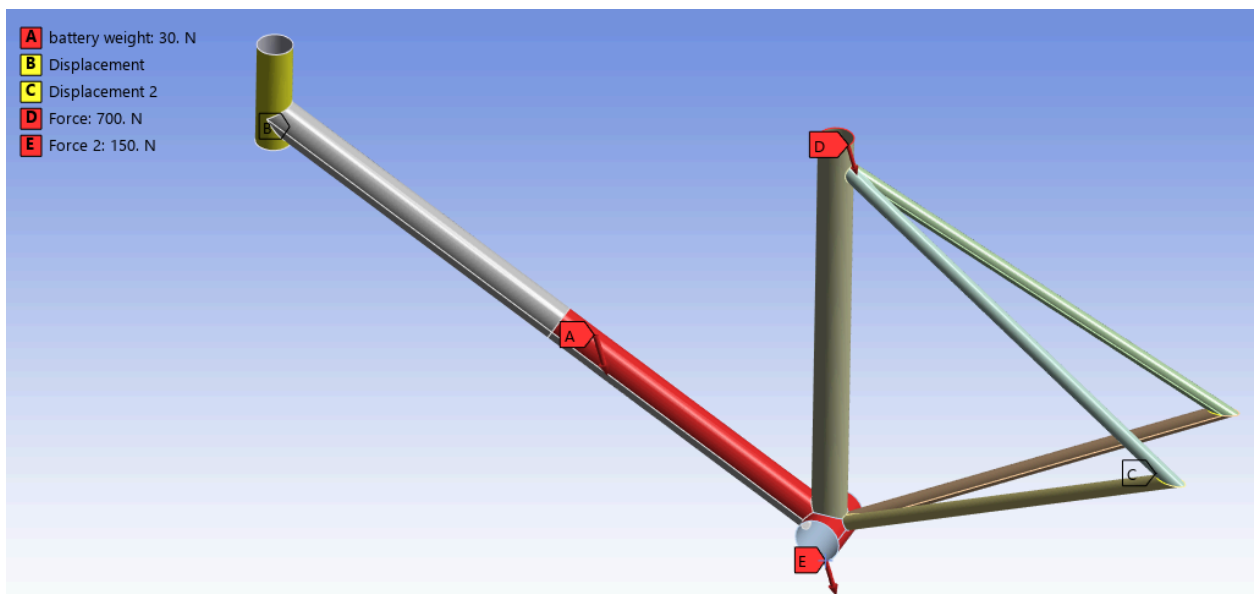


☐ Thickness	2. mm
Thickness Mode	Manual
Offset Type	Middle
Treatment	None
Material	
☐ Assignment	Aluminum Alloy

Statistics	
☐ Nodes	14080
☐ Elements	14086

Element Order	Linear
☐ Element Size	4.0 mm

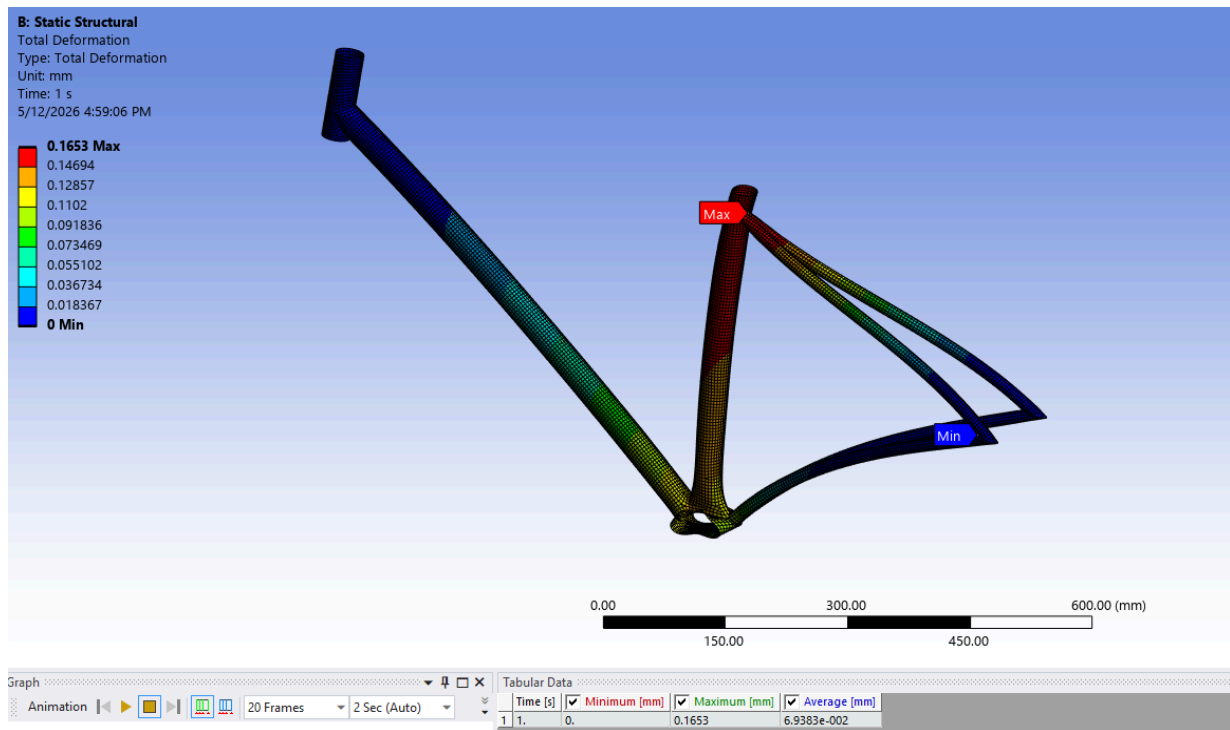
Boundary conditions and load was applied as below:

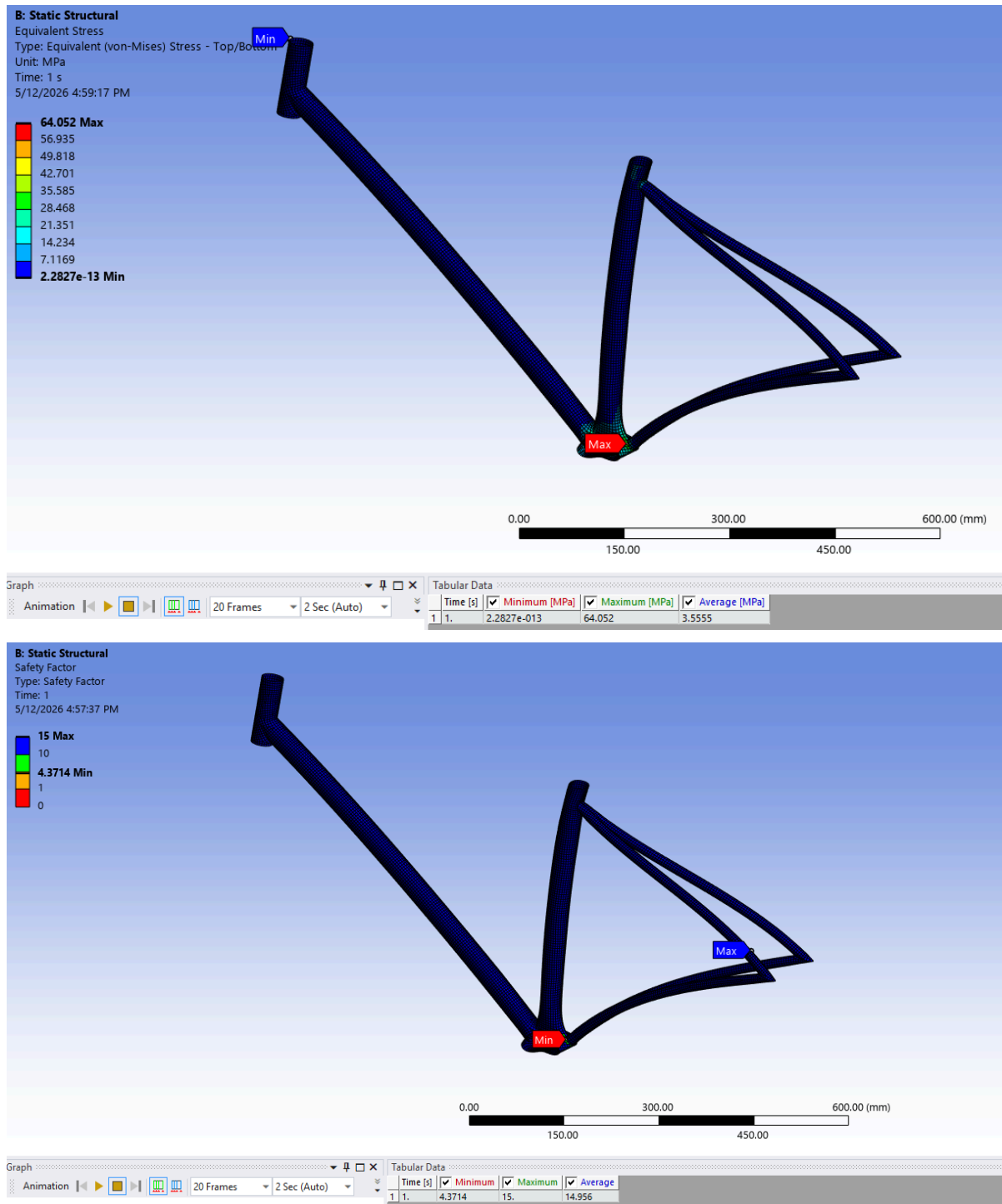


3.5 New Design Ansys Results Before Optimization

- Main stress location: Near the load transfer region at the top
- SF remained well above the required safety factor threshold

Result	Value
Total Deformation	Max 0.165 mm
Equivalent Stress	Max 64.1 MPa
Safety Factor	Min 4.4, Max: 15, Average: 14.96





The maximum deformation occurred near the seat tube/down tube region. The maximum deformation was only 0.165 mm, which is extremely small compared with the overall frame size of approximately 700 mm. This indicates that the new frame remained very stiff under the applied static loading conditions. The maximum equivalent stress was 64.052 MPa, located near the bottom bracket/seat tube region. This result is physically reasonable because the bottom bracket area is a major load-transfer region where multiple frame members meet.

4. Parametric Optimization

4.1 Optimization Objective

The optimization objective was to reduce mass while keeping the frame structurally safe. The design variable was the thickness of the tubes, to optimize them to min safety factor and weight reduction.

The project used response surface optimization. This method tests several design points, fits a response surface, predicts an optimal candidate, and then verifies that candidate with a direct FEA solve.

A design of experiments was generated in ANSYS to evaluate **different combinations of tube thickness** values. The design of experiments contained 82 points and took approximately 1.5h to solve. For each design point, ANSYS calculated the frame mass and minimum safety factor. These results were then used to build a response surface relating tube thicknesses to structural performance.

4.2 Design Variable

- Tube thickness being optimized
- Thickness range: 2–4 mm for all tube
- Output parameters: mass and safety factor

Parameter	Value
Optimized variable	Thickness of tubes
Lower bound	2 mm
Upper bound	4 mm
Objective	Minimize mass
Constraint	Safety factor ≥ 2
Outputs	Mass and minimum safety factor

Outline of Schematic B8: Parameters				
	A	B	C	D
1	ID	Parameter Name	Value	Unit
2	Input Parameters			
3	Static Structural (B1)			
4	P1	Body4 Thickness	2	mm
5	P4	Body20 Thickness	2	mm
6	P5	Body15 Thickness	2	mm
7	P6	Body18 Thickness	2	mm
8	P7	Body16 Thickness	2	mm
9	P8	Body17 Thickness	2	mm
10	P9	Body25 Thickness	2	mm
11	P10	Body27 Thickness	2	mm
*	New input parameter	New name	New expression	
13	Output Parameters			
14	Static Structural (B1)			
15	P2	Geometry Mass	1.1616	kg
16	P3	Safety Factor Minimum	4.3714	
*	New output parameter		New expression	
18	Charts			

4.3 Design of Experiments

- Input: Thickness of tube (2mm-4mm)
- Output: Mass for each design point
- Output: Safety factor for each design point

2	Design of Experiments	
3	Input Parameters	
4	Static Structural (B1)	
5	P1 - Body4 Thickness	☒
6	P4 - Body20 Thickness	☒
7	P5 - Body15 Thickness	☒
8	P6 - Body18 Thickness	☒
9	P7 - Body16 Thickness	☒
10	P8 - Body17 Thickness	☒
11	P9 - Body25 Thickness	☒
12	P10 - Body27 Thickness	☒
13	Output Parameters	
14	Static Structural (B1)	
15	P2 - Geometry Mass	
16	P3 - Safety Factor Minimum	
17	Charts	
18	Parameters Parallel	☒
19	Design Points vs Parameter	☒

Properties of Outline A5: P1 - Body4 Thickness		
	A	B
1	Property	Value
2	General	
3	Units	mm
4	Type	Design Variable
5	Classification	Continuous
6	Values	
7	Lower Bound	2
8	Upper Bound	4
9	Allowed Values	Any

4.4 Response Surface Results

The response surface allowed the design space to be explored more efficiently than manually testing each thickness combination. Instead of guessing which tubes should be thicker, the optimization compared mass and safety factor across the design range and identified candidate points that satisfied the safety factor requirement while reducing mass.

After a long long time of running..

Table of Outline A2: Design Points of Design of Experiments											
	A	B	C	D	E	F	G	H	I	J	K
1	Name	P1 - Body4 Thickness (mm)	P4 - Body20 Thickness (mm)	P5 - Body15 Thickness (mm)	P6 - Body18 Thickness (mm)	P7 - Body16 Thickness (mm)	P8 - Body17 Thickness (mm)	P9 - Body25 Thickness (mm)	P10 - Body27 Thickness (mm)	P2 - Geometry Mass (kg)	P3 - Safety Factor Minimum
44	43	3.4975	1.9005	1.9005	2.0995	2.0995	1.9005	1.9005	1.9005	1.4308	4.1508
45	44	2.5025	2.0995	1.9005	2.0995	2.0995	1.9005	2.0995	2.0995	1.2752	4.5653
46	45	3.4975	2.0995	1.9005	2.0995	2.0995	1.9005	2.0995	1.9005	1.4402	4.3944
47	46	2.5025	1.9005	2.0995	2.0995	2.0995	1.9005	2.0995	2.0995	1.2826	4.3129
48	47	3.4975	1.9005	2.0995	2.0995	2.0995	1.9005	2.0995	1.9005	1.4476	4.1497
49	48	2.5025	2.0995	2.0995	2.0995	2.0995	1.9005	1.9005	2.0995	1.2802	4.5641
50	49	3.4975	2.0995	2.0995	2.0995	2.0995	1.9005	1.9005	1.9005	1.4452	4.3931
51	50	2.5025	1.9005	1.9005	1.9005	1.9005	2.0995	2.0995	1.9005	1.2354	4.167
52	51	3.4975	1.9005	1.9005	1.9005	1.9005	2.0995	2.0995	2.0995	1.4513	4.4514
53	52	2.5025	2.0995	1.9005	1.9005	1.9005	2.0995	1.9005	1.9005	1.233	4.4184
54	53	3.4975	2.0995	1.9005	1.9005	1.9005	2.0995	1.9005	2.0995	1.449	4.7209
55	54	2.5025	1.9005	2.0995	1.9005	1.9005	2.0995	1.9005	1.9005	1.2403	4.1703
56	55	3.4975	1.9005	2.0995	1.9005	1.9005	2.0995	1.9005	2.0995	1.4563	4.455
57	56	2.5025	2.0995	2.0995	1.9005	1.9005	2.0995	2.0995	1.9005	1.2497	4.422
58	57	3.4975	2.0995	2.0995	1.9005	1.9005	2.0995	2.0995	2.0995	1.4657	4.7247
59	58	2.5025	1.9005	1.9005	2.0995	1.9005	2.0995	1.9005	2.0995	1.2654	4.5187
60	59	3.4975	1.9005	1.9005	2.0995	1.9005	2.0995	1.9005	1.9005	1.4304	4.3648
61	60	2.5025	2.0995	1.9005	2.0995	1.9005	2.0995	2.0995	2.0995	1.2748	4.7755
62	61	3.4975	2.0995	1.9005	2.0995	1.9005	2.0995	2.0995	1.9005	1.4398	4.6134
63	62	2.5025	1.9005	2.0995	2.0995	1.9005	2.0995	2.0995	2.0995	1.2821	4.5177
64	63	3.4975	1.9005	2.0995	2.0995	1.9005	2.0995	2.0995	1.9005	1.4471	4.3636
65	64	2.5025	2.0995	2.0995	2.0995	1.9005	2.0995	1.9005	2.0995	1.2798	4.7741
66	65	3.4975	2.0995	2.0995	2.0995	1.9005	2.0995	1.9005	1.9005	1.4448	4.612
67	66	2.5025	1.9005	1.9005	1.9005	2.0995	2.0995	1.9005	2.0995	1.2658	4.3879
68	67	3.4975	1.9005	1.9005	1.9005	2.0995	2.0995	1.9005	1.9005	1.4308	4.2283
69	68	2.5025	2.0995	1.9005	1.9005	2.0995	2.0995	2.0995	2.0995	1.2752	4.6524
70	69	3.4975	2.0995	1.9005	1.9005	2.0995	2.0995	2.0995	1.9005	1.4402	4.4845
71	70	2.5025	1.9005	2.0995	1.9005	2.0995	2.0995	2.0995	2.0995	1.2826	4.3915
72	71	3.4975	1.9005	2.0995	1.9005	2.0995	2.0995	2.0995	1.9005	1.4476	4.2317
73	72	2.5025	2.0995	2.0995	1.9005	2.0995	2.0995	1.9005	2.0995	1.2802	4.6562
74	73	3.4975	2.0995	2.0995	1.9005	2.0995	2.0995	1.9005	1.9005	1.4452	4.488
75	74	2.5025	1.9005	1.9005	2.0995	2.0995	2.0995	2.0995	1.9005	1.2566	4.3025
76	75	3.4975	1.9005	1.9005	2.0995	2.0995	2.0995	2.0995	2.0995	1.4726	4.5883
77	76	2.5025	2.0995	1.9005	2.0995	2.0995	2.0995	1.9005	1.9005	1.2543	4.5466
78	77	3.4975	2.0995	1.9005	2.0995	2.0995	2.0995	1.9005	2.0995	1.4703	4.8498
79	78	2.5025	1.9005	2.0995	2.0995	2.0995	2.0995	1.9005	1.9005	1.2616	4.3016
80	79	3.4975	1.9005	2.0995	2.0995	2.0995	2.0995	1.9005	2.0995	1.4776	4.5869
81	80	2.5025	2.0995	2.0995	2.0995	2.0995	2.0995	2.0995	1.9005	1.271	4.5454
82	81	3.4975	2.0995	2.0995	2.0995	2.0995	2.0995	2.0995	2.0995	1.487	4.8481

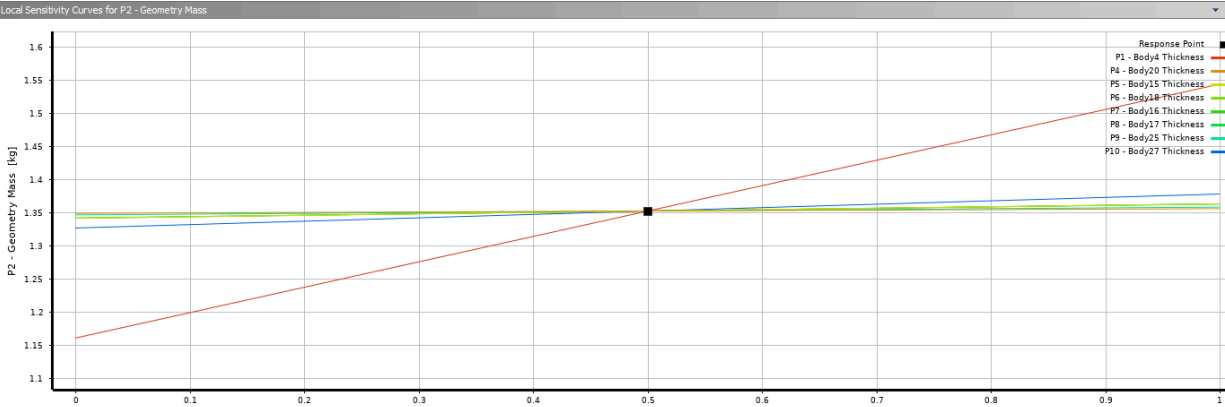
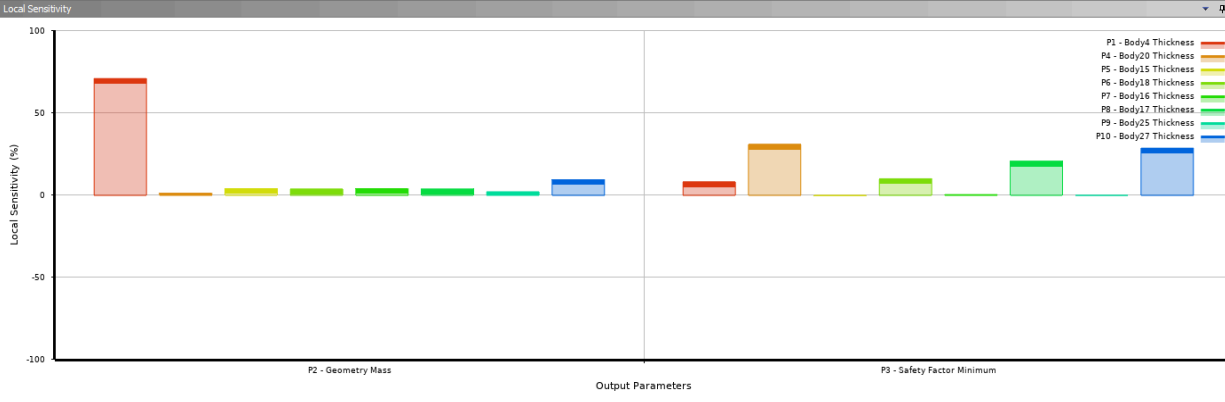
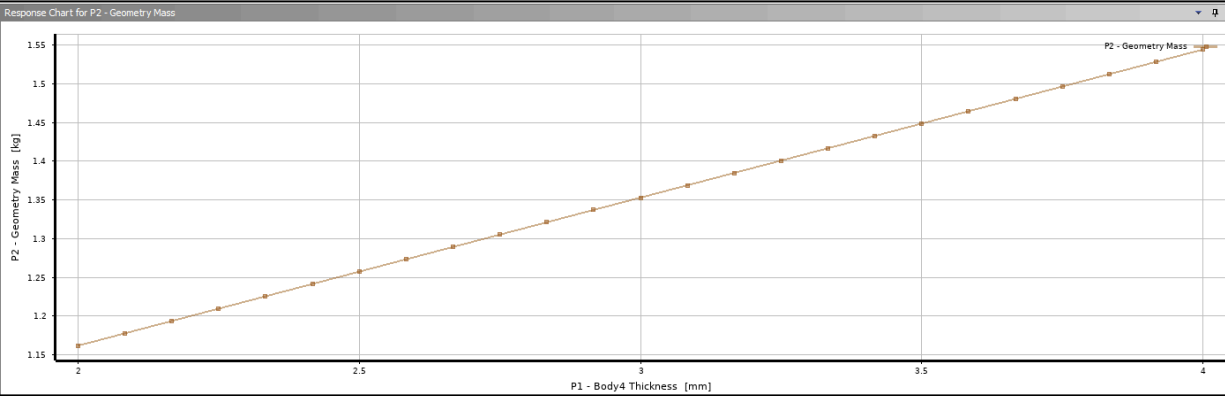
Response Surface: Min-Max Search

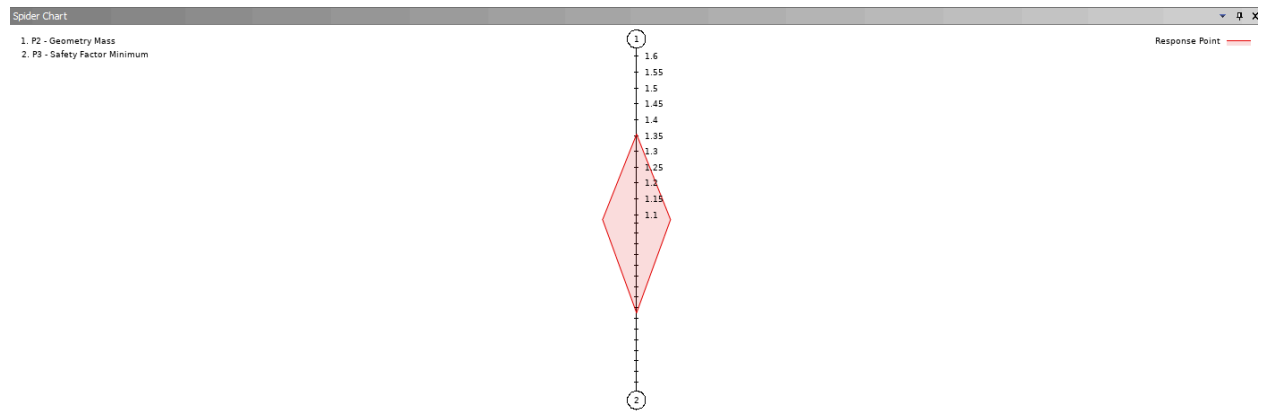
Table of Outline A17: Min-Max Search											
	A	B	C	D	E	F	G	H	I	J	K
1	Name	P1 - Body4 Thickness (mm)	P4 - Body20 Thickness (mm)	P5 - Body15 Thickness (mm)	P6 - Body18 Thickness (mm)	P7 - Body16 Thickness (mm)	P8 - Body17 Thickness (mm)	P9 - Body25 Thickness (mm)	P10 - Body27 Thickness (mm)	P2 - Geometry Mass (kg)	P3 - Safety Factor Minimum
2	Output Parameter Minimums										
3	P2 - Geometry Mass	2	2	2	2	2	2	2	2	1.1616	3.2186
4	P3 - Safety Factor Minimum	2	2	2.3342	2	2	2	2	2	1.1798	3.2181
5	Output Parameter Maximums										
6	P2 - Geometry Mass	4	4	4	4	4	4	4	4	2.3232	12.855
7	P3 - Safety Factor Minimum	4	4	4	4	3.5576	4	3.1368	4	2.2735	12.856

Response Surface: Tolerances

Table of Schematic D3: Response Surface: Tolerances					
	A	B	C	D	E
1	Name	Calculated Minimum	Calculated Maximum	Maximum Predicted Error	Refinement
2	P2 - Geometry Mass (kg)	1.1616	2.3232	2.4056E-15	☐
3	P3 - Safety Factor Minimum	3.2181	12.856	0.55976	☐

Table of Outline A25: Response Points											
	A	B	C	D	E	F	G	H	I	J	K
1	Name	P1 - Body4 Thickness (mm)	P4 - Body20 Thickness (mm)	P5 - Body15 Thickness (mm)	P6 - Body18 Thickness (mm)	P7 - Body16 Thickness (mm)	P8 - Body17 Thickness (mm)	P9 - Body25 Thickness (mm)	P10 - Body27 Thickness (mm)	P2 - Geometry Mass (kg)	P3 - Safety Factor Minimum
2	Response Point	3	2	2	2	2	2	2	2	1.353	4.4435
*	New Response Point										





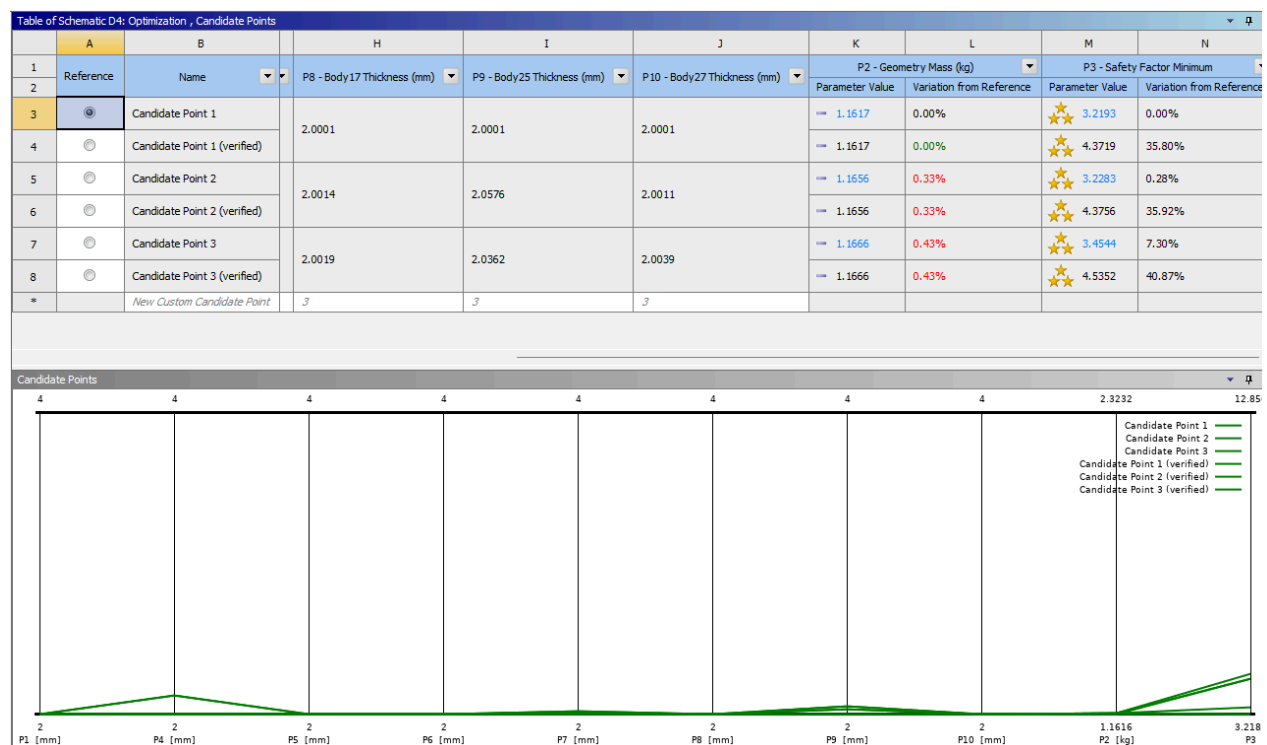
4.5 Verification of Optimized Candidate

Outline of Schematic D4: Optimization			
	A	B	C
1		Enabled	Monitoring
2	✓ Optimization		
3	Objectives and Constraints		
4	P3 ≥ 2		
5	Minimize P2		
6	Domain		
7	Static Structural (B1)		
8	P1 - Body4 Thickness	✓	
9	P4 - Body20 Thickness	✓	
10	P5 - Body15 Thickness	✓	
11	P6 - Body18 Thickness	✓	
12	P7 - Body16 Thickness	✓	
13	P8 - Body17 Thickness	✓	
14	P9 - Body25 Thickness	✓	
15	P10 - Body27 Thickness	✓	
16	Parameter Relationships		
17	✓ Convergence Criteria		
18	Results		
19	✓ Candidate Points		
20	✓ Tradeoff		
21	✓ Samples		
22	✓ Sensitivities		

Candidate Points!!!

	A	B	K	L	M	N
1	Reference	Name	P2 - Geometry Mass (kg)		P3 - Safety Factor Minimum	
2			Parameter Value	Variation from Reference	Parameter Value	Variation from Reference
3		Candidate Point 1	1.1617	0.00%	3.2193	0.00%
4		Candidate Point 1 (verified)	1.1617	0.00%	4.3719	35.80%
5		Candidate Point 2	1.1656	0.33%	3.2283	0.28%
6		Candidate Point 2 (verified)	1.1656	0.33%	4.3756	35.92%
7		Candidate Point 3	1.1666	0.43%	3.4544	7.30%
8		Candidate Point 3 (verified)	1.1666	0.43%	4.5352	40.87%
*		New Custom Candidate Point				

Candidate 1 had the lowest predicted mass, while Candidate 3 had the highest factor of safety. Since all three candidate points had predicted safety factors above 2, Candidate 1 was the best mass-minimizing design. However, Candidate 3 had the highest predicted safety factor with only a very small mass increase. For a bike-share frame, where durability and repeated use matter, Candidate 3 may also be attractive if a slightly higher mass is acceptable.



For the final optimized design, Candidate 1 seems to be the strongest choice, since it best matches the optimization objective of minimizing mass while keeping the safety factor above 2.

5. Reflection

5.1 What I Learned

This project helped me understand that finite element analysis is not just about creating the most realistic model. The most important part is deciding how to turn a real engineering problem into a simplified but meaningful simulation. For the bike frame, I had to think about what parts of the geometry mattered and how to simplify it, where the loads should be applied, what assumptions were reasonable, and how to interpret the results without over-trusting the software.

I also was able to see that stress results are more sensitive than deformation results. I realized I should be more cautious in the interpretation of it and really try to analyze how to best predict the stress and cross check it outside of ansys and with validation methods.

5.2 Challenges

One of the biggest challenges to figure out was setting up the model that was simple but still physically meaningful. While using simplified static loading and fixed boundary conditions, I had to think carefully about what the results actually meant and how to best simulate real life conditions.

Another challenge was interpreting stress results. Deformation stayed stable, but stress changed noticeably with element order and mesh changes. Like how there was a 20.45% increase when switching from linear to quadratic elements. This showed that stress needs more careful interpretation, especially near joints, load application points, and boundary conditions.

The optimization setup was also challenging because multiple tube thicknesses had to be managed as parameters. This required thinking about the design as a tradeoff rather than simply maximizing strength.

5.3 Future Improvements

If I had more time, I would improve the project in several ways. First, I would use more realistic boundary conditions instead of fully fixed supports and more realistic loading cases. Bike-share frames experience repeated loading, so fatigue and durability would be important for a real deployment.

Second, I would optimize more than one design parameter. In this project, the optimization focused on tube thickness, but future work could also optimize tube diameter, battery location, and joint geometry. This would give a better balance between mass, stiffness, safety factor, and manufacturability.

Finally, I would use submodeling near high-stress regions such as the bottom bracket and seat tube connection. These regions are important because they transfer loads between multiple frame members. A local submodel could help guide better reinforcement design.

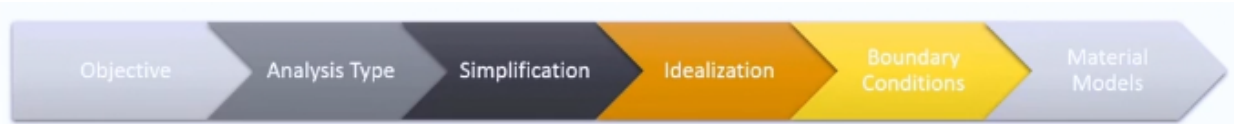
Overall, this project taught me that good simulation is not just about solving the model. It is about making thoughtful assumptions, checking whether the results make physical sense, and using the results to make better engineering decisions.

Appendix

Ansys Cross-Talk Summary

<https://innovationspace.ansys.com/certifications/thinking-like-an-analyst-ansys-crosstalk/>

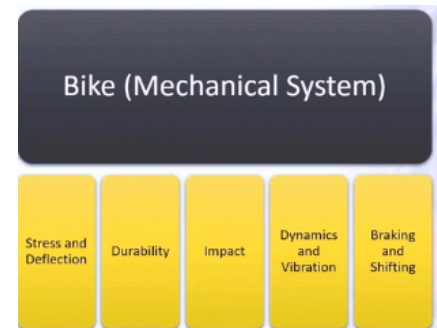
Main simulation modeling lessons (Think in steps as below)



- Understand your simulation objectives first! What are you trying to solve? And what is the best way to do so?
- Decide on how much detail to add

How engineering problem becomes FEA model

The biggest takeaway from the video was that good simulation is not about making the most complicated model possible. It is about building the simplest model that can still answer the engineering question. The simulation results are only meaningful if the assumptions behind the model are clear.



One of the key points mentioned was that the detail and real life resemblance should be determined based on the objective. If I'm just trying to understand where the frame has high stress, I don't need to include every tiny chamfer, fillet, or small CAD detail unless it actually changes the result. Adding too many small details can actually make the mesh way more complicated and in return slow down the simulation without really making the answer more useful.

I realized knowing when to and when not to use simplified models, knowing how in proximity to real life to model the system is a very important distinction an engineer should be able to see. This is both for the cad model being used, boundary conditions, loading situations and so on.

This connects directly to the bike frame project because the frame was modeled using shell elements. The shell model simplified the thin-walled tubes while still allowing the main stiffness, stress, and deformation behavior to be studied.

Another useful concept was idealization. A good analyst does not always model every connection with detailed contact. Depending on the goal, shared topology, simplified joints, beams, or other idealizations can represent the important structural behavior more efficiently. In this project, shared topology was useful because it allowed the frame tubes to behave as connected members while still allowing different tube thicknesses to be assigned for optimization.

Overall, the video helped me see that simulation is really about making smart engineering choices, not just making the model look realistic.

